GLENCORE

Asset Decommissioning & Disposal Management

11



Asset Decommissioning & Disposal Management

Operational Excellence, Glencore Copper
30 May 2022, Version 1.0

To close out the drafting of the manual a workshop was held with the following stakeholders in May 2022:

- Osvaldo Zamorano (Glencore Copper Asset Management – Metallurgical Assets)
- Nicolas Fortin (Horne-CCR)
- Yan Letourneau (Horne-CCR)
- Ikechukwu Iheanacho (CSA)
- Maximo Rojas (PASAR – *remote revision)*
- Nicholas Taylor (Glencore Copper commercial team - *remote revision)*
- Neale Laffin (Queensland Metals – Glencore Zinc)
- Robert Farrell (Murrin Murrin Operations – Glencore Nickel)
- Probin Konwar (Murrin Murrin Operations – Glencore Nickel)
- Sagar Karde (Murrin Murrin Operations – Glencore Nickel)
- Rohan Reisener (Glencore Copper Asset Management - Australia)
- Dino Moll (Glencore Copper Asset Management - Australia)
- Anthony Green (Glencore Copper Asset Management - APAC)
- Akeem Araoye - (Glencore Copper Asset Management – North America)
- Yuri Yurac - (Glencore Copper Asset Management – South America)
- Lawrence Hlatshwayo - (Glencore Copper Asset Management – Africa)

Supported by the AM Network Pty Ltd and Work Management Solutions Pty Ltd

Preface

The Asset Management Framework, Policy and Manuals are intended to provide leading Asset Management practices when applied and will be continually improved for the benefit of Glencore Copper.

The real value of sharing the knowledge gained from the use of these documents will come from continued discussion of the issues and rationale behind each document rather than by reading and interpreting them as individuals.

Readers are encouraged to consider these against our values around Safety, Responsibility, Simplicity, Integrity, Openness and Entrepreneurialism and how they reflect our Purpose – to responsibly source the commodities advancing everyday life. This document lays out the minimum standard for Glencore Copper and compliance to this manual will demonstrate a maturity value equal to Planned/Deployed (3) on the Glencore Copper Asset Management Maturity Scale.

Table of Contents

Preface 2
Table of Contents 3
Figures 3
Tables 3
1. Introduction 4
2. Objectives 4
3. Definitions 5
4. Approach 6
5. Deployment 13
6. Measures of Success 14
7. Improvement 14
8. References 15
9. Revision History 15
10. Appendix A – Example Disposal Questionnaire 16
11. Appendix A – Stakeholder Signoff 19

Figures

Figure 1-1 Asset Management Framework 4
Figure 4-1 Asset Decommissioning and Disposal Management 7

Tables

Table 4-1 Asset Decommissioning and Disposal Management Process Description 8
Table 4-2 Process RACI 10
Table 4-3 Functional Descriptions 11
Table 8-1 References 15

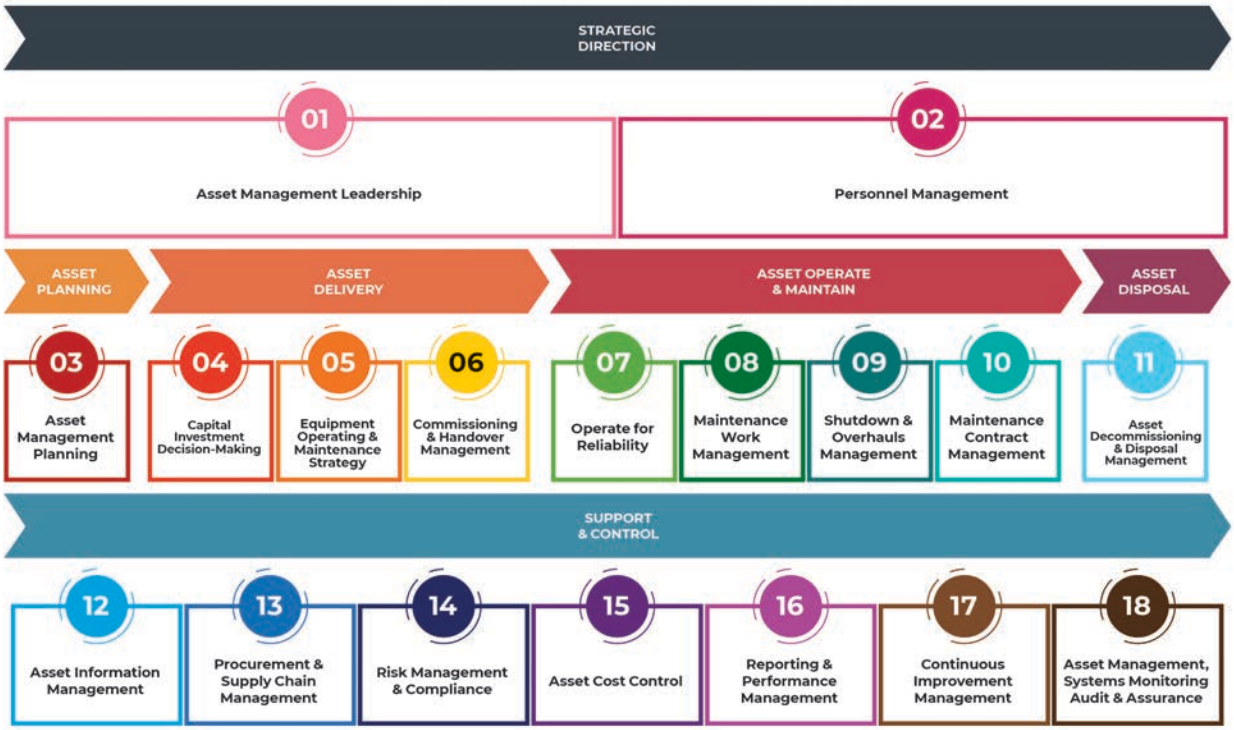
Copyright © Glencore 2022

1. Introduction

The Asset Management Policy and Manuals define how Glencore Copper manages the whole of life of an asset to ensure optimal value is extracted by ensuring assets are capable of meeting business and stakeholder objectives.

Figure 1-1 describes the high-level Asset Management Framework, which enables Glencore Copper to deliver its Asset Management purpose and vision.

FIGURE 1-1 ASSET MANAGEMENT FRAMEWORK



2. Objectives

The objective of this manual for Asset Decommissioning and Disposal Management is to ensure compliance with internal and regulatory requirements.
This objective will be accomplished through the following drivers:

- Meeting legislative obligations for the decommissioning and disposal of assets.
- All systems supporting management of assets in operation are amended to reflect asset decommission decisions.
- The decision to decommission assets are viewed as an opportunity to review lifecycle performance.
- Glencore Copper procedures for asset decommissioning and disposal are met.
- Managing the transfer of assets from one site to another to maintain asset integrity, commercial requirements and legal obligations, on an as needs basis.

Copyright © Glencore 2022

2.1 Accountabilities

Dedicated resources must be assigned, and capabilities developed within each Operation to deliver Asset Decommissioning and Disposal Management process that enable effective delivery of the Asset Management objectives.

Assign accountability for each or all the following:

- Compliance with statutes, legislation and Glencore Copper requirements when decommissioning and disposing of assets
- The decommissioning of assets
- Storage requirements for decommissioned assets
- Decommissioned assets are disposed of, transferred or sold as required.

2.2 Scope

This manual is to be read in conjunction with the following:

- Capital Management Framework (CMF)
- Glencore Corporate Risk Framework
- Element E01 Asset Management Leadership
- Element E02 Personnel Management
- Element E03 Asset Management Planning
- Element E05 Equipment Operating and Maintenance Strategy
- Element E13 Procurement and Supply Chain
- Element E14 Risk Management and Compliance
- Element E15 Asset Cost Control.

The scope includes ensuring:

- Clear authorities and responsibilities for decommissioning and disposal of assets are understood and managed.
- Asset Management systems show the correct status of decommissioned assets.
- Commercial requirements managed for disposal of assets including sale, transfer or discarding of assets.
- Legal obligations are managed for disposal of assets.
- Asset Integrity is managed for stored assets.
- Health, Safety, Environment and Community risks are understood and well managed.

3. Definitions

Decommissioning – actions required to remove an asset from service.

Disposal – actions necessary to decommission and dispose of assets that are no longer required.

Residual Value - the estimated value of an asset at the end of its lease term or useful life.

Useful Life - is typically referred to as the length of time an asset generates revenue for a company or adds value to the owner.

Copyright © Glencore 2022

4. Approach

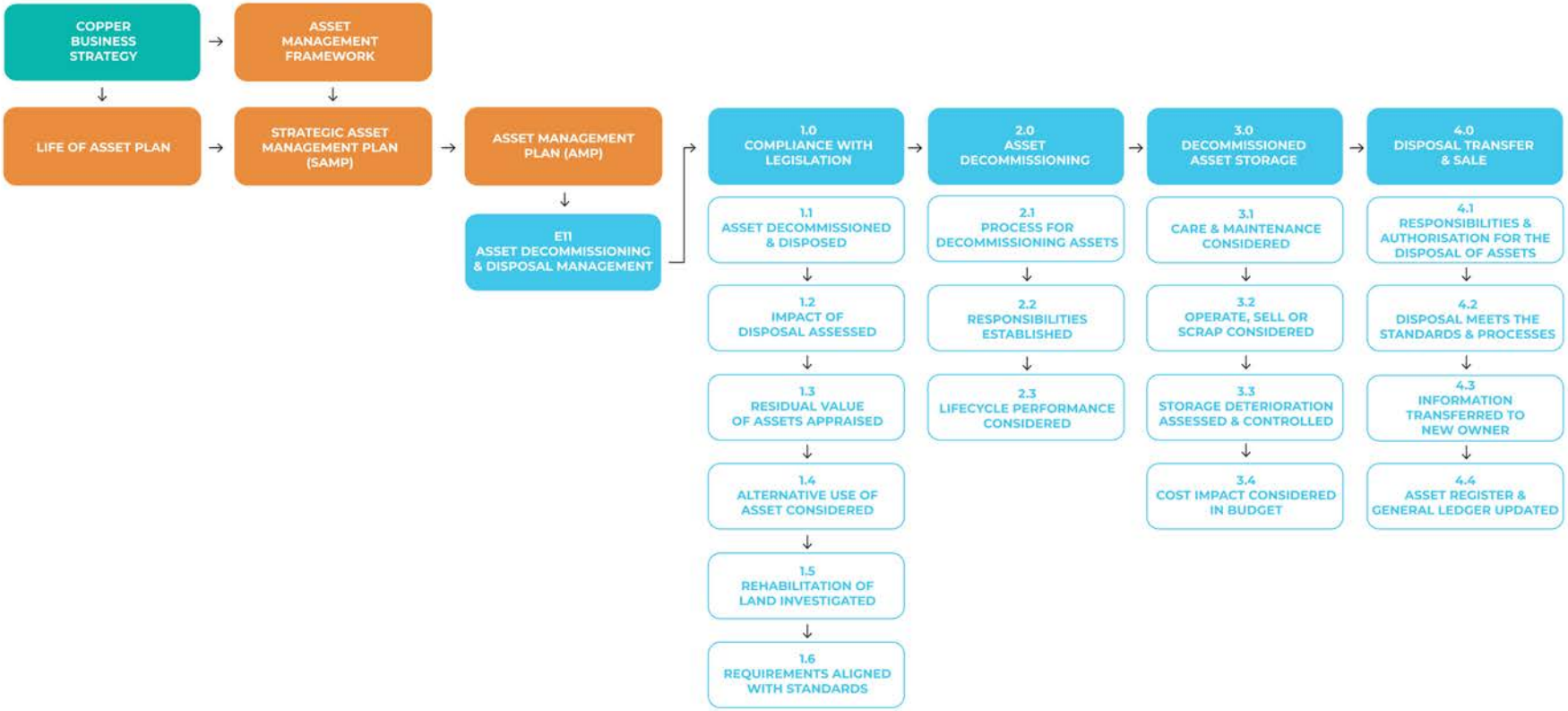
The approach for Asset Decommissioning and Disposal Management is described through:

- Process Flow - pictorially illustrates the key Asset Decommissioning and Disposal Management activities, see Figure 4-1.
- Process Description – defines in more detail the key Asset Decommissioning and Disposal Management activities, see Section 4.2.
- Process RACI Matrix – indicates for each activity, the Functions who are responsible, Accountable, who are Consulted prior to final decision/action and those who are Informed of any outcomes and actions, see Section 4.3.

Copyright © Glencore 2022

4.1 Process Flow

FIGURE 4-1 ASSET DECOMMISSIONING AND DISPOSAL MANAGEMENT



Copyright © Glencore 2022

4.2 Process Description

TABLE 4-1 ASSET DECOMMISSIONING AND DISPOSAL MANAGEMENT PROCESS DESCRIPTION

PROCESS STEP	DESCRIPTION
1.0 Compliance with Legislation	**Developing processes for assets that address legislative requirements for the decommissioning and/or disposal of individual assets through to site/mine closure.**
1.1 Assets Decommissioned and Disposed	Decommissioning and disposal need the same focus as other lifecycle activities. Asset disposal requirements are laid out in the Asset Management Plan for the asset and shall be consulted to ensure that assets are decommissioned and disposed of in accordance with the plan. A Decommissioning and Disposal Plan is to be developed.
1.2 Impact of Disposal Assessed	When disposing of assets, environmental impact is to be taken into consideration and management of any hazardous material and waste.
1.3 Residual Value of Assets Appraised	If an asset is to be sold, then the residual value of the assets needs to be assessed. The residual value of the assets is calculated as: Residual Value = (estimated salvage value) – (cost of asset disposal)
1.4 Alternative use of Asset Considered	If an asset is not going to be sold, then consideration is to be given to see if the asset can be used in an alternative role. For example, a loader which is at end of life could be used for clearing and other activities if it not economically viable to dispose of. If a site closure is to occur, then consideration is to be given to transfer the asset/s to another location if useful life still remains.
1.5 Rehabilitation of Land Investigated	If land is to be rehabilitated then it must be done as per the legal and statutory requirements of the country and region, including any decontamination requirements.
1.6 Requirements Aligned with Standards	Glencore Coppers mine/site closure planning requirements are to be followed and compliance to Glencore Copper Environmental Policy when mines/site is to be closed.
2.0 Asset Decommissioning	**Developing and implementing at site a process for decommissioning assets.**
2.1 Process for Decommissioning Assets	In addition to this manual for Asset Decommissioning and Disposal, sites shall develop a detailed process for decommissioning of assets at site level. The process shall cover at a minimum, the following: • Updating the Enterprise Resource Planning (ERP) status for the assets, including removing from service, transfer to other areas or sites. • Updating the Planned Maintenance schedule, including turning off maintenance plans, transfer of data if required and update of Asset Management Plans and budgets. • Updating spare parts inventory and the consideration for inventory transfer to other regions/sites, or inventory disposal, whether scraped or sold. • Reviewing and updating any contracts that are in place for the assets such as supply, repair and overhaul. • The development of risk management plans aligned to the Glencore Copper Risk Management Framework and Element E14 Risk Management and Compliance manual. • Ensuring that competent people are used in the development of and delivery of the decommissioning process. • Ensuring that safe work procedures are developed, implemented and managed.
2.2 Responsibilities Established	Refer to Element E02 Personnel Management. Responsibilities for asset decommissioning are to be established and position descriptions are to reflect these responsibilities.
2.3 Lifecycle Performance Considered	The site decommissioning process shall take into consideration lifecycle performance when making decisions when assets are to be decommissioned.

Copyright © Glencore 2022

	Refer to Element E03 Asset Management Planning which provides a process by which assets can be assessed for condition, remaining life and next major replacement/overhaul activity. This information can be used for decision making purposes regarding when assets are required to be decommissioned.
3.0 Decommissioned Asset Storage	**Developing and implementing arrangements for the storage and 'care and maintenance' of decommissioned assets to ensure tolerable risk levels for HSEC and any future recommissioning requirements.**
3.1 Care and Maintenance Considered	Should assets still have determined residual life remaining then consideration is to be given for care and maintenance of those assets while in storage. Master data is to be set up/modified in the CMMS to ensure that there is a formal process by which care and maintenance is to be executed and is to follow the requirements of E08 Maintenance Work Management.
3.2 Operate, Sell or Scrap Considered	Once decommissioned, consideration is to be given regarding the management of that asset, such as: • Continue to operate – if the asset can be used in alternative systems/locations then consideration is to be given for the transfer of those assets. • Sell – if the asset has financial commercial value, then consideration for selling the asset is to be given. • Scrap – if the asset has been assessed as end of life and has no financial commercial value then consideration to scrap the asset is to be given.
3.3 Storage Deterioration Assessed and Controlled	If decommissioned assets are to be stored, then the risks associated with deterioration in storage are to be assessed and actions put in place to reduce the risks of deterioration. Consider E05 Equipment Operating and Maintenance Strategies through this process.
3.4 Cost Impacts Considered in Budget	The cost impacts for decommissioned assets are to be established as part of the budgeting process. Cost impacts should include: • Preservation – work required to prevent the asset from deteriorating • Care and maintenance • Warehousing • Statutory and legal requirements • Valuation – to establish the market value if sold.
4.0 Disposal, Transfer and Sale	**Ensuring the disposal, transfer or sale of assets is approved by an authorised person and conducted in line with a defined process, which confirms compliance with all Glencore Copper corporate and legislative requirements.**
4.1 Responsibilities and authorisation for the Disposal of Assets.	Responsibilities are to be given to authorised person that are able to dispose of, transfer or sell assets. Authorisation is required prior to the disposal, transfer or sale of any assets. Refer to Glencore Copper Delegation of Authority (DoA) process.
4.2 Disposal Meets the Standards and Processes	When disposing of assets all statutory, legal and Glencore Copper disposal requirements are to be met. Consider Asset Management Plans where documented. An example disposal questionnaire can be seen at Appendix A.
4.3 Information Transferred to New Owner	If the asset is to be transferred or sold, all the asset information may be transferred to the new owner including: • Information regarding statutory and legal compliance • Maintenance records • Spare parts records • Data sheets • Risk assessments • Modification records • Financial records.

Copyright © Glencore 2022

4.4 Asset Register and General Ledger Updated	Once the asset has been disposed, transferred or sold the asset and financial register are to be updated to reflect the removal of the asset from service and the financial register.

4.3 Process RACI Description

The RACI matrix below in Table 4-2 addresses the high-level functional elements of the Asset Decommissioning and Disposal Management process. For the overall Asset Management RACI matrix, refer to the Asset Management Intranet page.

Note: The HSE and Commercial Departments do not appear in this high-level RACI, although their involvement will be required from time to time dependent on the nature of the variance.

The codes used in the RACI analysis are defined as follows:

TABLE 4-2 PROCESS RACI

R	Responsible	The "doer" – the individual(s) who will perform an activity or directs others to perform the activity.
A	Accountable	"The buck stops here" –the individual who ultimately "owns" the activity – there should only be one person.
C	Consulted	Person that needs to feedback and contribute to the activity.
I	Inform	The individual(s) who needs to be informed after a decision or action is taken if relevant. (via 1-way communication).

Function / Process Step	General Management	Maintenance Management	Operations Management	Maintenance Execution	Supply / Warehouse	Maintenance Planning	Technical Services	Environmental	Finance
1.0 Compliance with Legislation									
1.1 Assets Decommissioned and Disposed		A	C	R	R		R	C	C
1.2 Impact of Disposal Assessed		A	C	C			C	R	
1.3 Residual Value of Assets Appraised		A		C			C		R
1.4 Alternative use of Asset Considered		A	C	R	C	C	R		C
1.5 Rehabilitation of Land Investigated	A	C	C				C	R	
1.6 Requirements Aligned with Standards	A	R	R	C			R	R	R
2.0 Asset Decommissioning									
2.1 Process for Decommissioning Assets		A	R	R		C	R	C	C
2.2 Responsibilities Established		A	R	R		C	R	C	C
2.3 Lifecycle Performance Considered		A	C	C	C	C	R		C
3.0 Decommissioned Asset Storage									
3.1 Care and Maintenance Considered		A	R	R	R	C	C		
3.2 Operate, Sell or Scrap Considered		A	R	R	C		R	C	C

Copyright © Glencore 2022

Process Step \ Function	General Management	Maintenance Management	Operations Management	Maintenance Execution	Supply / Warehouse	Maintenance Planning	Technical Services	Environmental	Finance
3.3 Storage Deterioration Assessed		A		R	R	C	R		
3.4 Cost Impact in Budget		A		R	R	C	C	C	
4.0 Disposal, Transfer and Sale									
4.1 Responsibilities and authorisation for the Disposal of Assets.	A	R	C	C	R	C	C	C	C
4.2 Disposal Meets the Standards and Processes		A	C	C	R	C	C	C	C
4.3 Information Transferred to New Owner		A	C	C	R	C	C	C	C
4.4 Asset Register and General Ledger Updated		A	R		R	R			R

TABLE 4-3 FUNCTIONAL DESCRIPTIONS

Function	Description
General Management	• Ensures asset decommissioning and disposal is conducted in line with statutory, legal and environmental requirements. • Ensures responsibilities are assigned for the disposal, transfer and sale of decommissioned assets.
Maintenance Management	• Ensures asset decommissioning and disposal plan is developed and implemented • Ensures any environmental impact for asset disposal in considered together with hazardous material management. • Ensures assets that are to be sold have residual value calculated • Ensures decommissioned assets are assessed to see if there is an alternative use of the asset. • Responsible for following the Glencore Copper planning and environmental requirements when mines/sites are closed • Provides input into land rehabilitation post site/mine closure. • Ensures a process for asset decommissioning is established and responsibilities are assigned. • Ensures lifecycle performance is considered in the asset decommissioning process • Ensures storage requirements for decommissioned assets are implemented. • Ensures decommissioned assets are, disposed of, transferred or sold. • Ensures storage costs are included in the budget process. • Ensures all statutory and legal requirements are met with regards to asset disposal. Also ensures that Glencore Copper disposal process if followed. • Ensures the allocation of responsibilities for the transfer, sale and disposal of decommissioned assets. • Ensures asset information is transferred to the new owner and the asset and financial registers are updated.
Maintenance Execution	• Responsible for the development and implementation of the asset decommissioning and disposal plan. • Provides input into the environmental impact of the decommission and disposal of assets. • Responsible for the consideration of alternative use of decommissioned assets. • Responsible for assessing decommissioned assets for alternative use.

Copyright © Glencore 2022

	• Provides input into ensuring asset decommissioning and disposal conforms to statutory and legal requirements. • Responsible for the establishment of a process for asset decommissioning is established and responsibilities are assigned. • Provides input into the consideration of lifecycle performance in the asset decommissioning process • Responsible for ensuring that decommissioned assets are stored appropriately. • Responsible for the care and maintenance of decommissioned assets. • Provides input into the disposal of, transfer and sale of decommissioned assets.
Maintenance Planning	• Provides input regarding assessment of decommissioned assets for alternatives use. • Provides input into the asset decommissioning process and decommissioning of assets. • Provides input into the care and maintenance of decommissioned assets, storage deterioration prevention and lifecycle performance consideration. • Provides input for the disposal of, transfer and sale of decommissioned assets.
Operations Management	• Provides input regarding asset decommissioning process and asset disposal. • Provides input into the alternative use of decommissioned assets. • Provides input into the investigation regarding the rehabilitation of land. • Responsible for following the Glencore Copper planning and environmental requirements when mines/sites are closed. • Ensures a process for asset decommissioning is established and responsibilities are assigned. • Provides input into the consideration of lifecycle performance in the asset decommissioning process • Provides input into the care and maintenance of decommissioned assets and decisions regarding operating, selling or scrapping decommissioned assets. • Provides input for the disposal of, transfer and sale of decommissioned assets.
Supply/Warehouse	• Responsible for the development of asset decommissioning and disposal plan. • Provides input into the alternative use of decommissioned assets. • Provides input into the development of the asset decommissioning process. • Responsible for the care and maintenance of decommissioned assets. • Provides input regarding asset operate, sell or scrap process. • Responsible for ensuring that decommissioned assets are stored appropriately. • Responsible for the disposal, transfer and sale of decommissioned assets.
Technical Services – Mining, Processing, Engineering	• Responsible for the development and implementation of the asset decommissioning and disposal plan. • Provides input into the environmental impact of the decommission and disposal of assets. • Provides input in appraising assets residual value and the rehabilitation of land investigations. • Responsible for the consideration of alternative use of decommissioned assets and that all requirements are aligned to Glencore Copper's standards. • Responsible for the decommissioning of assets. • Responsible for assessing decommissioned assets for alternative use. • Provides input into ensuring asset decommissioning and disposal conforms to statutory and legal requirements. • Responsible for the establishment of a process for asset decommissioning is established and responsibilities are assigned. • Responsible for lifecycle performance in the asset decommissioning process • Responsible for ensuring that decommissioned assets are stored appropriately. • Provide input to the care and maintenance of decommissioned assets. • Provides input into the disposal of, transfer and sale of decommissioned assets.

Copyright © Glencore 2022

Environmental	• Provides input with regards to asset decommissioning and disposal • Responsible with respect to appraising the residual value of assets. • Responsible for following the Glencore Copper planning and environmental requirements when mines/sites are closed • Provides input regarding assessment of decommissioned assets for alternatives use. • Provides input with regards to asset decommissioning • Provides input with regards to the consideration to operate, sell or scrap decommissioned assets. • Provides input with respect to the budget cost impact of decommissioning assets. • Provides input with regards to the disposal, transfer and sale of assets.
Finance	• Provides input with regards to asset decommissioning and disposal • Responsible with respect to the appraisal of asset residual value • Responsible for following the Glencore Copper planning and environmental requirements when mines/sites are closed • Provides input regarding assessment of decommissioned assets for alternatives use. • Provides input into the asset decommissioning process and decommissioning of assets. • Provides input with regards to the consideration to operate, sell or scrap decommissioned assets. • Provides input with regards to the disposal, transfer and sale of assets.

5. Deployment

The deployment of the Asset Decommissioning and Disposal Management approach across all business areas is through:

- Awareness presentation
- Position description alignment
- Competency assessment
- Training, coaching and mentoring.

5.1 Awareness Presentation

This Manual is supported by a slide presentation and video to help facilitate communication and to raise awareness on how it contributes to the Asset Management Framework and business success.

5.2 Position Description Alignment

To effectively deploy this Manual, the essential process steps and activities need to be diligently mapped and assigned to approved positions within Glencore Copper.

The RACI matrix assists in the visualisation that everyone is working towards the same goal. It clarifies, not only the individual's responsibilities, but also that of others. It has proven to be an effective tool to quickly understand and close the gaps between current and desired practices and business process activities.

Position descriptions including KPIs need to be aligned to the RACI matrix and measures of success. This is the basis for ongoing performance reviews and the creation and execution of individual development plans.

Copyright © Glencore 2022

5.3 Competency Assessment

This Manual is supported by an Asset Management competencies assessment tool which is used to:

- Guide recruitment, selection, development and deployment of people in asset management roles across the company
- Provide employees responsible for managing assets with feedback, development, coaching and recognition for their performance.
- Map, build and manage the asset management talent pool across the company.
- Identify strengths and development needs in asset management practitioners.

5.4 Training, Coaching and Mentoring

Coaching and mentoring are another way to assist employees improve their understanding of their role and to improve performance.

Coaching is generally about improving performance in the short term where mentoring is about achieving high performance and career aspirations over a longer term.

Coaching or mentoring requires addressing that a person is not currently up to the required level to perform their role as effectively and efficiently as they could. The build-up of skills to perform efficiently and effectively requires a large investment of time and cooperation.

Coaches and Mentors need to be carefully selected and must be committed for the long term to effect maximum success.

6. Measures of Success

To ensure success of the implementation of this manual the following should be considered:

- Ensure all assets that are disposed are flagged accordingly in the CMMS and they are moved to the graveyard area of the CMMS.
- Ensure a disposal check sheet/questionnaire is developed and linked to the disposed asset.
- Ensure there is 100% compliance to the disposal plan.

7. Improvement

The process to review and continually improve the approach and deployment for Asset Decommissioning and Disposal Management is scheduled Annually, or when there is a sudden change in the Operating Context. As the organization matures a longer review period could be considered.

Copyright © Glencore 2022

8. References

The following documents provide detailed support material that should be read in conjunction with this Manual to provide a comprehensive understanding of Asset Decommissioning and Disposal management.

TABLE 8-1 REFERENCES

DOCUMENT TITLE	DESCRIPTION
Asset Management Framework	This document defines the overall direction of asset management across Glencore Copper.
Glencore Copper Capital Management Framework	Details the Capital Management requirements for Glencore Copper laid out in three parts for minor, intermediate and major capital projects.
Institute of Asset Management SSG 6&7 Capital Investment, Operations and Maintenance Decision Making	Provides guidance regarding Capital Investment Decision Making.

9. Revision History

VERSION NO.	REASON FOR REVISION	DATE
0.1	New Manual to support the Asset Management Framework	18/01/2022
0.2	Reformat and Issue for Final Review	18/05/2022
1.0	Update Site Signoff and Publish	30/05/2022

Copyright © Glencore 2022

10. Appendix A – Example Disposal Questionnaire

Question 1	What Type of Disposal? (see the list of layout types A to F and their description below).
Question 2	Has the SD (Sustainable Development) list been completed taking into account the disposal of this asset(s)? If not, please comment on the risks in the comment box.
Question 3	Your approval is required for the disposal of these assets (approval is required by three stakeholders: Process Engineer (owner of the equipment), Head of Prevention Hygiene (for environmental risks, Chief Engineer (must assess the risk for the company)
Question 4	Have you analysed the effect of the inactivity of this asset(s)? (this question is asked of the electrical and mechanical maintenance engineer) if the type of arrangement is D or E
Question 5	Should a PM (periodic review) of this asset(s) be created? If yes, indicate your recommendations in the comment box and contact the planner. If not, please comment why a PM is not required in the comment box. (this question is asked of the electrical and mechanical maintenance engineer) if the type of arrangement is D or E
Question 6	Make adjustments based on the type of layout and feedback received from the maintenance engineer. Have you created a PM to cover the General Inspectorate? If yes, please indicate in the comment box on which No Installation the costs will be charged since no fees can be charged against this asset. (this question is asked of the electrical and mechanical planner) if the type of arrangement is D or E
Question 7	Have you entered the disposal value in $ of this asset(s) to be disposed of in your comment area and have you had the disposal process validated by the department head? If this has not started, you are required to begin managing the sale or disposal of this asset(s). This question is asked of the buyer.
Question 8	Enter your instructions (if required) for the planner in the comment box (this question is asked of the electrical and mechanical maintenance engineer)
Question 9	If there is a pressure vessel among the items here, have you notified the head of utility services? (this question is asked of the electrical and mechanical maintenance engineer)
Question 10	Are there any service contracts related to this asset(s) that need to be changed? (this question is asked of the electrical and mechanical maintenance engineer)
Question 11	"If the maintenance engineer has indicated that changes to the service contract for this asset(s) are to be made in question 10, then: Have you made the changes to the service contract? (MSO220: Cancel equipment maintenance contract). (this question is asked of the buyer)"
Question 12	Have you updated the high-level equipment registry? (this question is asked of the electrical and mechanical maintenance engineer)
Question 13	Have you analysed the removal and validation of OPS as needed with the process engineer? If you choose Not Applicable, please indicate the reason in the comment box. (this question is asked of the electrical and mechanical maintenance engineer)
Question 14A	Have you idled maintenance plans? (this question is asked of the electrical and mechanical planner)

Copyright © Glencore 2022

Question 14B	For layout type A, B or C: Have you closed Work Orders (IW32) as well as process Maintenance Notifications (IW22)? For type of disposal D or E: Have you put Work Orders on Hold status (HOLD)? (this question is asked of the electrical and mechanical planner)
Question 14C	Have you removed the Equipment Ranges (IA02) link? (this question is asked of the electrical and mechanical planner)
Question 15A	Is this asset a store item? (this question is asked of the electrical and mechanical planner)
Question 15B	Are there any store items related to this asset? (this question is asked of the electrical and mechanical planner)
Question 15C	Have you idled maintenance plans? (this question is asked of the electrical and mechanical planner)
Question 15D	Have you analysed whether the store item is required on other equipment? (this question is asked of the electrical and mechanical planner)
Question 15E	Have you consulted with all staff who may be affected by the disposal of this store item(s) from inventory? (this question is asked of the electrical and mechanical planner)
Question 15F	Have you reviewed the BOM IE03 and sent an FM-09 to the store supervisor? If you chose Not Applicable, please explain why in the comment box. (this question is asked of the electrical and mechanical planner)
Question 15G	Have you modified the MDG (modify parts list) to take into account the elements to be modified by the planner of the other discipline? If you chose Not Applicable, please explain why in the comment box. (this question is asked of the electrical and mechanical planner)
Question 16A	Have you looked at the history of this store item(s) to make sure it is not used elsewhere? (For cases where this item(s) store is not connected to equipment). If you chose Not Applicable, please explain why in the comment box. (Based on the analysis of the FM-09 received, this question is asked of the Store Supervisor)
Question 16B	Have you removed this item(s) store or adjusted the quantity (MMBE Migo-551)? If you chose Not Applicable, please explain why in the comment box. (Based on the analysis of the FM-09 received, this question is asked of the Store Supervisor)
Question 16C	Have you indicated the total value of the store items in the comment box? Keep in mind that adjustments to the value of inventories must be authorized in accordance with section 6.42 of the DAM. (Based on the analysis of the FM-09 received, this question is asked of the Store Supervisor)
Question 17A	Has PLC programming been removed? (this question is asked of the automation engineer)
Question 17B	Has UIO programming been removed? (this question is asked of the automation engineer)
Question 17C	Have the PI points been removed? (this question is asked of the automation engineer)
Question 17D	Has the technical manual been updated? (this question is asked of the automation engineer)
Question 17E	Have the network components been updated? (this question is asked of the automation engineer)
Question 18	Have you corrected SAP to reflect the status of this asset(s) (NWBC EAM Governance Home Page) (and other databases if applicable)? (This question is asked to the Equipment Manager; in our case it is Milena Dimitrova)
Question 19A	Have you removed this asset(s) from the lockout sheets? (This question is asked to prevention)
Question 19B	Will this asset disposal affect the confined space inventory? (This question is asked to prevention)
Question 19C	Will this asset disposal affect the inventory of nuclear sources? (This question is asked to prevention)
Question 19D	Will this asset provision affect the Inventory of Hazardous Materials Used in the Workplace (WHMIS)? (This question is asked to prevention)

Copyright © Glencore 2022

Question 19E	Will this asset provision affect machine safety inventory? (This question is asked to prevention)
Question 19F	Will this asset disposal affect the content of safe work requirements (TSEs)? (This question is asked to prevention)
Question 19G	Will this asset disposal affect the absence of the asset create an unsafe work area (risk of falling)? If yes, notify the supervisor responsible for the area to correct the situation. (This question is asked to prevention)
Question 20	Have you assessed the impact of the disposal of this asset(s) on the property tax bill? (This question is put to the Accounting Officer management-asset)
Question 21	Have you obtained the cost of acquiring this asset(s)? (This question is put to the Accounting Officer management-asset)
Question 22	Have you obtained the accumulated value of the depreciation of this asset(s)? (This question is put to the Accounting Officer management-asset)
Question 23	Have you entered the book value of this asset(s)? (This question is put to the Accounting Officer management-asset)
Question 24	Have you completed Glencore's asset disposal form? (This question is put to the Accounting Officer management-asset)
Question 25	Have you had the approval of the director general and other directors? (This question is put to the Accounting Officer management-asset)
Question 26	Have you notified the person in charge of licence plate renewal if it is a mobile device (MBL)? (This question is put to the Accounting Officer management-asset)
Question 27	Have you updated the capital asset registry? (This question is put to the Accounting Officer management-asset)
Question 28	Have you affected the ledger? (This question is put to the Accounting Officer management-asset)

Types of Disposal offered:	**Note: An Asset is used to represent an installation (equipment), a productive unit, a store item.**
A – Relocation of an asset:	When an asset is relocated, the No Installation could be renamed, the production unit and the account code could be changed. This includes an internal sale of Xstrata Copper.
B- Sale of an asset:	Includes donations, add a note in the comment box. Does not include an internal sale of Xstrata Copper, see A.
C- Disposal of an asset:	Total elimination of assets.
D- Long-term storage of an asset:	When an Asset is inactive and stored in a different place from its usual location. Becomes the responsibility of the store.
E- Inactivity of an asset (Standby /Out of service):	When an asset remains in place but is no longer used. All maintenance programs must be removed and if general maintenance is to be applied, it will have to be loaded onto the building or other No Installation.
F- Removal of item(s) store (stock code):	To formalize the withdrawal of important store item(s). Can also be used to formalize the withdrawal of item(s) store that should have been removed in the past.

Copyright © Glencore 2022

11. Appendix A – Stakeholder Signoff

Stakeholder Name	Site or Department	Contact	Stakeholder Signoff
Anthony Green	Asset Management - Australia	Anthony.green@glencore.com.au	7/6/2022
Akeem Araoye	Asset Management – North America	Akeem.Araoye@glencore.ca	7/6/2022
Yuri Yurac	Asset Management – South America	Yuri.yurac@glencore.cl	8/6/2022
Lawrence Hlatshwayo	Asset Management – Africa	Lawrence.hlatshwayo@glencore.co.za	10/6/2022
Rohan Reisener	Asset Management - Australia	Rohan.reisener@glencore.com.au	7/6/2022
Dino Moll	Asset Management - Australia	Dino.Moll@glencore.com.au	13/6/2022

Disclaimer

Copyright © Glencore 2022

This document is the property of Glencore and contains information that is the Intellectual Property of Glencore or is otherwise proprietary to Glencore. This document and the information contained herein shall not be made available to any person or persons not authorised to receive it nor shall it be reproduced, copied, loaned, exhibited or used for any other purpose whatsoever, in whole or in part, without the prior written consent of Glencore.

Copyright © Glencore 2022

Notes

Copyright © Glencore 2022

Notes

Copyright © Glencore 2022

Notes

Copyright © Glencore 2022

Asset Decommissioning & Disposal Management

11



Asset Decommissioning & Disposal Management

Rohan Reisener
Manager – Asset Management Strategy and Development
Glencore Copper Operational Excellence
T: +61 7 3833 8408 M: +61 478 316 195
rohan.reisener@glencore.com.au
160 Ann St, Brisbane QLD 4000